

Sanidhya Gupta

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RESEARCH INTERESTS

Distributed quantum computing, quantum error correction, quantum networking, modular quantum architectures, and measurement-based quantum computation.

EDUCATION

Durham University, United Kingdom

Oct. 2025 – Present

Ph.D. in Computer Science (Quantum Information and Networking)

- Supervisors: [Dr. Thirupathaiah Vasantam](#) and [Prof. Neil Walton](#)

Indian Institute of Science Education and Research (IISER) Bhopal, India Aug. 2019 – June 2024

BS-MS, Major in Electrical Engineering and Computer Science; Minor in Data Science

- CPI: 9.01/10.00

PUBLICATIONS & PREPRINTS

- S. Gupta**, S. Bhambay, N. Alavisamani, N. Walton, and T. Vasantam (2026). *Boundary-Aware Stabilizer Scheduling for Distributed Quantum Error Correction*. Submitted to **IEEE International Conference on Quantum Computing and Engineering (QCE)**. [arXiv:2604.22471](#).
- S. Gupta** and A. Raina (2025). *A Provably Secure Framework for Noise-Aware Delegated Quantum Computation and Storage*. Under review at **Quantum Information Processing**. [arXiv:2403.07596](#).
- S. Gupta** and A. Raina (2024). *A feed-forward method for encoding a Stabilizer Code using Measurement-Based Quantum Computing*. QADA Workshop at **IEEE QCE**, Montréal, Canada. DOI: [10.1109/QCE60285.2024.10252](#).
- C. Pandey, **S. Gupta**, R. R. Das, and A. Raina (2023). *Quantum Network Coding and Distribution of Maximally Entangled States in Measurement-Based Quantum Computing*. **National Conference on Communications (NCC)**, IIT Guwahati, India. DOI: [10.1109/NCC56989.2023.10067927](#).

SELECTED PRESENTATIONS

- Workshop presentation: *A feed-forward method for encoding a Stabilizer Code using Measurement-Based Quantum Computing*. QADA Workshop at **IEEE QCE**, Montréal, Canada, 2024.
- Poster presentation: *A feed-forward method for encoding a Stabilizer Code using Measurement-Based Quantum Computing*. **International Conference on Quantum Communication, Measurement and Computing (QCMC)**, IIT Madras, India, Aug. 26–30, 2024.

FELLOWSHIPS

- EPSRC Doctoral Landscape Award (2025 – 2029)**: Fully funded doctoral scholarship awarded by the UK Engineering and Physical Sciences Research Council (UKRI) to support postgraduate research at Durham University.
- Chanakya PG Fellowship (2024)**: Selected as one of only 9 recipients nationwide. Awarded by the Department of Science and Technology, Government of India, for postgraduate research in quantum information science.
- Chanakya UG Fellowship (2022)**: Selected as one of only 5 recipients nationwide. Awarded by the Government of India for undergraduate research in quantum information science.

RESEARCH EXPERIENCE

Ph.D. Researcher | Durham University

Oct. 2025 – Present

Supervisors: Dr. Thirupathaiah Vasantam and Prof. Neil Walton

- Scheduling for distributed quantum error correction**: Developing seam-refresh policies for modular QEC over noisy photonic interconnects, with circuit-level simulation in Stim and analysis of logical-error behavior under entanglement-generation delays.
- Magic-state architectures for distributed quantum computing**: Analysing centralised magic-state supply memory architectures using queueing models, blocking probability, and waiting-time analysis for modular quantum systems.

Junior Research Fellow | IISER Bhopal

Apr. 2025 – Jul. 2025

Supervisor: Dr. Ankur Raina

- Led development of a prototype quantum-network simulator and coordinated the work of research interns.
- Guided implementation of QKD protocols including BB84, COW, and DPS under configurable hardware parameters.

MS Thesis Researcher | IISER Bhopal

May 2023 – Sep. 2024

Error-resilient Distributed Quantum Computation | Supervisor: Dr. Ankur Raina

- Developed stabilizer-code-based protocols for noise-aware delegated quantum computation and storage.
- Designed MBQC-based stabilizer encoding gadgets, leading to a QCE 2024 workshop paper and a QCMC 2024 poster presentation.
- Completed thesis research on secure and distributed quantum computation using measurement-based techniques.

Undergraduate Researcher | IISER Bhopal

May 2022 – Feb. 2023

Quantum Network Protocols using MBQC | Supervisor: Dr. Ankur Raina

- Developed an MBQC-based protocol for quantum network coding and entanglement distribution.
- Validated the protocol analytically and numerically, resulting in a publication at NCC 2023.

AWARDS AND HONORS

- **QHack 2024:** Team ranked 17th among 700+ global teams.
- **IBM Quantum Challenge:** Recognized among top scorers globally in Spring 2021; also participated in Fall 2021 and Spring 2023.
- **IBM Certified Associate Developer:** Quantum Computation using Qiskit v0.2X.

TEACHING, MENTORSHIP AND SERVICE

- **Teaching Assistant, NPTEL Workshop:** Hands-on Programming for Quantum Computers (2024, 2025).
- **Teaching Assistant, Introduction to Quantum Computer Science (AY 2023–24-I):** supported tutorials, laboratory sessions, and examinations.
- Led interactive quantum-computing demonstrations for high-school and college students during IISER Bhopal Open House (2022).

TECHNICAL CAPABILITIES

- **Quantum:** Stim, Qiskit, Cirq, PennyLane, TensorFlow Quantum, Amazon Braket
- **Programming:** Python, C/C++, MATLAB, CUDA
- **Scientific Computing:** NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib
- **Platforms:** IBM Quantum Cloud, AWS

WORKSHOPS AND SUMMER SCHOOLS

- **QCom(P) Workshop at COMSNETS 2024:** focused on quantum communication and quantum computing systems.
- **Qiskit Global Summer School (2021, 2022, 2023):** training in quantum computing, simulation, and quantum machine learning.
- **Quantum Computing and Programming (QCourse511), 2021:** online course by the University of Latvia and QWorld, including a hands-on project using TensorFlow Quantum.